PORTFOLIO

BAYU AJI TRISASONGKO

Electrical Engineering Graduate | Mechanical Design, Electronics, Mechatronics & Robotics

Bringing mechanical design, electronics, and control together to turn engineering ideas into tangible prototypes. An electrical engineering graduate with a solid hands-on foundation in mechatronics, component fabrication, and rapid prototyping. Eager to contribute and learn within engineering teams developing robotics, automation, and precision mechanical systems.

View Projects ↗ | Curriculum Vitae | Email | LinkedIn | WhatsApp

SCAN KODE QR ATAU KLIK DI MANA SAJA UNTUK BUKA PORTOFOLIO WEB
https://portobayu.vercel.app/
Bayu Aji Trisasongko · S1 Teknik Elektro, Universitas Telkom



Telkom University
S1 Teknik Elektro (2026)

Mechanical
3D CAD modeling, mechanism design, and fabrication.

Electronics
Schematic design, circuit wiring, and hardware assembly.

Embedded Control
Microcontrollers, sensors, and actuator integration.

Prototyping
Physical build, functional testing, and validation workflows.

01 / FINAL PROJECT



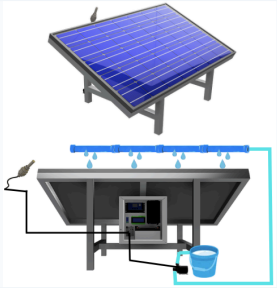
Robot Manipulator 5-DOF ↗

5-DOF articulated robotic arm prototype powered by an STM32F411 microcontroller and 3D-printed cycloidal gearboxes. Integrates AS5600 magnetic encoders in a closed loop to reduce backlash, achieving 79.8% accuracy across 250–1,000 g loads.

STM32F411 MCU · NEMA 17 Stepper · TB6600 Driver · AS5600 Sensor · TCA9548A Mux · Cycloidal Gearbox

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02 / CAPSTONE DESIGN



Pendingin Panel Surya Otomatis ↗

Automated water spray cooling system to lower solar panel surface temperature driven by Arduino Uno. Uses temperature sensing and relay control to trigger a submersible pump above >30°C and shut down below <25°C to restore efficiency.

Arduino Uno R3 · RTD Temp Sensor · 5V Relay Module · 12V Water Pump · 16x2 I2C LCD · 12V Battery Pack

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03 / KONTES ROBOT ABU INDONESIA (KRAI)



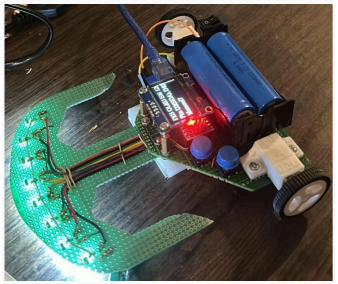
Kompetisi Robot Basket ↗

Mobile basketball robot for Kontes Robot ABU Indonesia (KRAI). Served as a core mechanical team member designing the shooter, dribbler, and drive wheel configuration, advancing to the National Stage at ITB Jatinangor.

6-Bar Pneumatics · 24V Solenoid · RS775 DC Motor · Omniwheel Drive · Limit Switch & IR · Air Regulator

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04 / PROYEK MANDIRI



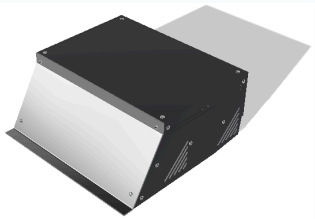
Robot Line Follower ↗

Autonomous line-tracking robot featuring differential drive steering. Equipped with an 8-sensor bar built from LED and photodiode pairs, an Arduino Nano microcontroller, a TB6612FNG driver, and micro N20 metal gear motors.

Photodiode Array · Arduino Nano · TB6612 Driver · N20 Gear Motor · 0.96" I2C OLED · 2S Li-ion Battery

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05 / PROYEK MANDIRI



Robot Sumo ↗

1 kg competition-class sumo robot (20 × 20 cm) built on an acrylic chassis with a front aluminum scoop. Delivers 4 kg push force via 4 PG24 planetary gear motors, steered wirelessly from a smartphone via Bluetooth connection.

Arduino Uno · 4x PG24 Motors · L298N Driver · HC-05 Bluetooth · 3S LiPo Battery · Acrylic Chassis

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